CA1 - Atlantic Renewable Care

1. Scenario Interpretation:

1.1: Organisation & Current Process

Atlantic Renewable Care maintains residential, commercial, and agricultural solar, wind, and power storage modules. It performs inspections, repairs, and maintenance on systems, some of which were installed by other companies. Information on customers, installations, visits, quotations, et cetera are scattered throughout different organisational formats both physical and digital, electronic and analog, though mechanical formats are highly unlikely.

Installation components can be made by different manufacturers, installations are made of multiple components, sites can have multiple different installations, and customers can own multiple sites. Component and installation information is recorded by staff if possible, but may be missing. Services can be ordered during business hours, or scheduled to repeat, and require varying visits and effort from the technicians. There is no reliable method for office staff to ascertain the completion status of a job. The methods of accessing installations per site should only be accessible to relevant technicians. These methods include personal contacts, passcodes, or equipment.

Service faults are reported by phone, email, and online form, and can concern performance, indicators, auditory and visual anomalies, and complete installation failure. Problems may be diagnosed by direct inspection following customer description, which office staff will record, and may request additional information, which will be analysed by a tech lead to determine the appropriate course of action. Emergency service is not provided, though urgent issues can be prioritised and may require customer intervention. Problems and reports have a many-to-many relationship.

Technicians have varying skills, experience, and authorisations. A single job can require multiple specializations, technicians, access methods, and equipment. Scheduling is affected by proximity and weather. Components are held in storage, ordered when needed, and provided by manufacturers under warranty. Visits may be scheduled without confirmation that the required components are in stock. Repairs can be temporary, additional work can be identified, and sites can be inaccessible or unsafe. Customer approval of a quotation is required to begin non-covered work. Approval can be provided by phone. Methods of recording customer approval differ between employees. Optional items may be included in quotations. Quotations can change after inspection, but must not change after work concludes.

Observations, readings, work completion, components used, and photographs taken are recorded by technicians during or after their tasks using various levels of technology, leading to some complete reports remaining unfiled or undelivered. A visit won't always fix the target fault, and may find additional faults that will require additional customer approval. Resumption of normal service requires inspection and approval by an authorised specialist. Whether the technician who performed the repairs or maintenance should be permitted to perform the inspection has yet to be decided within the organisation. Customers need to be able to access a report of the work done without gaining access to sensitive information.

The service manager requires information regarding installations and work done regarding them, inventory, and technician workload. Customers desire progress updates

without unverified information or conclusions. Technicians electronic devices can lose connection in rural areas. Office staff should have access to schedules and customer communications. Technicians require relevant site and installation information for their assignments. The proposed scope covers maintenance coordination and records. The proposed scope does not cover live energy monitoring, equipment control and customer billing systems.

1.2: Problems, Consequences, & Desired Outcomes

Customer and job information is spread across multiple formats, and delivery of records and updates can be delayed, inconsistent, or incomplete. Records of work performed can be delayed or fail to reach office staff, and may fail to convey whether the target fault was handled. This causes organisational dysfunction and inhibits staff from formulating a consistent view of the situation, and the information must be consolidated into a single format.

Sensitive information must be hidden from unauthorised employees and customers, customer update reports must refrain from including unconfirmed conclusions, and management needs access to scheduling, employee, and workflow-related data. This slows productivity and risks legal liability, and appropriate permission-based information-gatekeeping measures should be implemented.

Customer fault reports identify symptoms, not faults. Any number of reports may point to a single fault, multiple faults, or no faults at all. Fault priority designation fails to follow an established structural hierarchy, as is the case with safety-related concerns. Return-to-service designation authorisation is yet to be decided. This may hamper productivity and a detailed ruleset should be established.

Visits may be booked before the required inventory to solve the issues is confirmed to be in stock, and may fail to account for weather, travel, and site conditions. Office staff may fail to be notified of completed work due to reporting delays, and visits may not result in a resolved fault. This may cause confusion and should be corrected where possible.

Urgent work may result in components becoming unavailable or consumed without inventory being updated, leading to repairs becoming impossible to complete during the scheduled visit. Documentation of component transport should be ensured by mandating inventory movement be performed reliably.

Recording of quotation approval is inconsistent. Inspection may reveal additional faults that have not been weighed in pricing or service agreement, which itself can be unclear to staff. Breach of contract or unauthorised work can result in legal action. Strict rulesets must be established and a list of work actions should be created.

1.3: Known Evidence & Uncertainty

Statement	Classification	Reason
Information on customers, installations, visits, quotations, et cetera are scattered throughout different organisational formats.	Problem Evidence.	I got it from the scenario.

Statement	Classification	Reason
Information must be consolidated into a single format.	Inference.	It would alleviate issues with the above problem.
Existing records can be modified to fit a single format.	Assumption.	I'm pretty nifty with relational databases.
Fault priority assignment doesn't follow a ruleset.	Problem Evidence.	Tad inconvenient.
Urgent work may result in components becoming unavailable or consumed without inventory being updated.	Problem Evidence.	Can't do a job without the necessary parts.
Customer approval of a quotation is required to begin non-covered work.	Problem Evidence.	Obtaining customer approval is finicky.
Quotations can change after inspection, but must not change after work concludes.	Assumption.	I assume the company is using contracts.
Office staff may fail to be notified of completed work due to reporting delays.	Problem Evidence.	I still don't know what reason I'm supposed to be giving.
Visits may not result in a resolved fault.	Problem Evidence.	Kind of obvious.
Return-to-service designation authorisation is yet to be decided.	Problem Evidence.	Pretty big oversight.

2. Purpose, Scope, & Boundary:

2.1: Purpose

The system will, at the very least, provide a central hub for the company's records.

2.2: Scope

In Scope	Out of Scope
Customer, site, & installation reports.	Installation monitoring.
Symptom & fault reports, & investigation findings.	Installation control.
Maintenance visits & their results.	Customer billing.

In Scope	Out of Scope
Technician schedules, skills, authorisations, & jobs.	
Component inventory, utility, shipments, and reassignment.	
Quotations & customer approval thereof.	
Return to service designation ruleset.	
Role-based access privileges.	
Offline synchronisation.	

2.3: Dependencies & Constraints

The company needs to standardise company procedures, review records for consistency, and designate roles for access control.

3. Stakeholder & Actor Catalogue:

3.1: Stakeholders

Stakeholder	Type(s)	Goals	Concerns	Influence
Service Manager	Managerial stakeholder Primary actor	Plans Visits	That records about jobs, as well as technician schedules, will be inadequate.	Yes
Technician	Technical stakeholder Primary actor	Performs technical work	That records of jobs or sites will be inadequate. That required components or equipment will be unavailable.	Yes
Office Staff	Direct user Primary actor	Manages records, schedules, and communications	That records will fail to be delivered or will be inconsistent. That customer approval will fail to be recorded or will be inadequate.	Yes

Stakeholder	Type(s)	Goals	Concerns	Influence
Customer	Direct user Primary actor	Reports symptoms and faults	That work updates will be delayed, incomplete, or unreliable. That there will be additional charges not in the original quotation. That there will be unauthorised work done.	Yes
Technical Lead	Technical / managerial stakeholder Primary actor	Review inspections	That assessments will be incomplete or inadequate.	Yes
Parts Coordinator	Managerial stakeholder Primary actor	Manage inventory	That inventory records will be incomplete or inadequate.	Yes
Owner(s)	Governance stakeholder Offstage stakeholder	Make money	That they won't make money	Yes
Manufacturer(s)	External stakeholder Supporting actor	Supply parts	That their components will break before warranty expires	No

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3.2: Questions

- What rules should fault priority assignment follow?
- What technician skillset is required for which job?
- Who can permit an installation to resume normal service?

4. Requirements Catalogue:

4.1: Functional Requirements

ID	Functional Requirement	MoSCoW	Source/Rationale
FR0	The system shall provide authorised staff a means to update and link records.	Must	Office staff / needs an updated record of company operations

ID	Functional Requirement	MoSCoW	Source/Rationale
FR1	The system shall provide a means to report faults and symptom information.	Must	Customer / Fault reports are needed to justify scheduling a visit outside of regular maintenance.
FR2	The system shall assign work only to technicians with the required skillsets.	Must	Technician / Technicians need to be capable of completing the work they're assigned.
FR3	The system shall provide a means to record visits, the technicians assigned to them, and the progress thereof.	Must	Service Manager / Workflow needs to be tracked to ensure productivity.
FR4	The system shall provide a means to record installation condition updates, measurements, as well as maintenance and repair procedures.	Must	Technical Lead / Inspection, repair, and maintenance records are required for quality assurance and resumption of normal service decisions.
FR5	The system shall provide a means to assess inventory levels, availability of required components, and component warranties.	Must	Parts Coordinator / Accurate logistical records are necessary to function at peak efficiency.
FR6	The system shall provide a means to record and access customer approval of quotations and work.	Must	Office Staff / This is required for legal reasons.
FR7	The system shall provide a list of requirements for installations to resume normal service.	Must	Technical Lead / Someone needs a standardised ruleset for assessing whether an installation can resume normal service.
FR8	The system shall provide a means to publish and access progress reports and verified conclusions.	Should	Customer / Customers would like to know how their work is coming along and whether or not it has been completed.

ID	Functional Requirement	MoSCoW	Source/Rationale
FR9	The system shall provide a means to save reports when offline and submit when connection is restored.	Should	Technician / This would allow for reports to be made during/ immediately following a visit, whilst memory remains.

4.2: Non-Functional Requirements

ID	Quality	Measurable Requirement	Rationale
NFR0	Security	All attempts to access unauthorised information will be denied.	Sensitive information must be appropriately hidden.
NFR1	Data Integrity	Once a record has been saved, it will be available upon each subsequent system access.	Deleting records is typically a bad idea, especially with regulators down our throats.
NFR2	Offline Usability	Reports submitted offline will automatically be submitted once connection is restored.	It's convenient and can prevent procrastination.
NFR3	Performance	Most record queries should complete in under 5 seconds.	I don't know how big the company is but that should be plenty of time for even a massive company.

5. Rules & Uncertainty Register:

5.1: Business Rules

Category	Statement
Business Rule	Non-covered work must not commence until explicit customer approval is acquired.
Business Rule	Component reassignment must be mirrored in the inventory database.
Business Rule	Installations must not resume normal service before required checks are completed.

Category	Statement
Business Rule	Technicians must only be assigned work they are qualified for.
Assumption	Current spreadsheets and paper records can be consolidated into a single database.
Assumption	Staff can be restricted or granted access to different information based on their role.
Constraint	System scope excludes customer billing and installation monitoring and control.
Constraint	Visits may take place in areas without reliable internet reception.
Constraint	Repair and maintenance conclusions must not be conflated with approval to resume normal service.
Risk	Current records are incomplete, duplicated, or inconsistent.
Risk	Offline submissions conflict during synchronisation.
Risk	Sensitive information is leaked to unauthorised persons.

5.2: Clarification Questions

- How are fault priority, resumption of normal service permission, and customer approval determined?
- Which technicians' skills are required for each job or task?
- Who can approve resumption of normal service?
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6. Use-Case Model:

6.1: System Boundary

The system organises customer, installation, component, visit, site, report, quotation, technician, inventory, and scheduling records into a single database. It excludes customer billing and installation monitoring and control.

6.2: Actor Catalogue

See 3.1: Stakeholders

6.3: Use-Case Catalogue

ID	Use Case	Primary Actor	Meaningful Outcome	Source Requirement
UC0	Migrate, update, and maintain past records and add new ones.	Office Staff	All previous records are consolidated into the database and integrated with newer data.	FR0
UC1	Report fault symptoms.	Customer	Symptom reports are stored in the database, linked to the installation, and the technical lead is notified.	FR1
UC2	Investigate fault reports.	Technical Lead	Reports are investigated and results are recorded.	FR1, FR4
UC3	Schedule visits.	Service Manager	work is assigned to technicians with the necessary skills.	FR2, FR3
UC4	Record visit results.	Technician	Proper visit documentation is produced.	FR3, FR4
UC5	Inventory management.	Parts Coordinator	Component amounts, assignments, reassignments, decommission, and warranties are kept up to date.	FR5
UC6	Produce a quotation.	Office Staff	Potential work is priced.	FR6
UC7	Approve quotation.	Customer	Quotation is approved or rejected.	FR6
UC8	Approve resumption of normal service	Technical Lead	Perform necessary checks and approve or deny resumption of normal service.	FR7

ID	Use Case	Primary Actor	Meaningful Outcome	Source Requirement
UC9	Organise workloads and review work done.	Service Manager	Create balanced work schedules and ensure work is done properly.	FR3

6.4: User Stories

ID	Who	What	Why
US0	Customer	View progress reports	Ensure work is pregressing
	As a **customer** I want to **view reliable progress reports** so that I can **know that work is progressing smoothly**.		
US1	Office Staff	Receive quotation approval	Authorise quoted work to begin
	As **office staff**, I want to **receive provable quotation approval** so that **it can be documented and work can begin**.		
US2	Technician	Receive site access	access sites
	As a **technician**, I want to **be able to view site access protocol** so that I can **get in, do my job, and leave**.		

6.5: Detailed Use-Case

See 6.3: UC7

6.5.1: Preconditions

1. Installation is recorded in the system.
2. A symptom report is recorded in the system.
3. A technician has assessed the installation.
4. A fault was found and recorded during the assessment.
5. The fault report has been assessed and found to be valid.
6. A quotation for the proposed work has been drafted.

6.5.2: Trigger

Drafted quotation is sent to the customer and is approved.

6.5.3: Alternatives

4: No fault was found during the assessment.
5: The fault report has been assessed and found to be a false positive.
Trigger: Drafted quotation is sent to the customer and is rejected.

6.5.4: Exceptions
1, 2, & Trigger: National telecommunications infrastructure has collapsed and both access to the system and communications with the customer are currently impossible.
Trigger: Drafted quotation is sent to the customer but the customer has died since prior communication.

6.5.5: Postconditions
The quotation is approved and work can begin.

7. Gherkin Examples:
See /features/ca1-gherkins.feature

8. Candidate-Concept Analysis:

8.1: Candidate Terms

Candidate Term	Evidence Source	Initial Observation
Customer	Scenario	A customer cannot be owned.
Site	Scenario	A customer can own multiple sites.
Access Method	Scenario	Site access may be restricted by one or more access methods.
Installation	Scenario	A site may contain multiple installations.
Component	Scenario	May possess a warranty.
Manufacturer	Scenario	Create and supply components.
Symptom Report	Scenario	Symptoms may underlie a fault.
Fault	Scenario	A symptom report may lead to the discovery of zero to many faults.
Assessment	Scenario	Faults are diagnosed and service-readiness are determined via assessmen
Visit	Scenario	Work is performed during a visit.
Technician	Scenario	Visits are conducted by at least one technician.
Skill	Scenario	Technicians possess various skills.
Equipment	Scenario	Technicians conduct their work using equipment.
Quotation	Scenario	Cost of work is summarised in a quotation.
Approval	Scenario	Quotations are approved by customers.

Candidate Term	Evidence Source	Initial Observation
Work	Scenario	Actions taken by technicians during visits are generally classified as work.
Work Report	Scenario	Work performed during visits are summarised in work reports.
Inventory	Scenario	Stored and reserved components are tracked in inventory.
Assignment	Scenario	Components can be reserved for specific installations.
Reassignment	Scenario	Component assignment can be changed to solve more urgent faults.
Warranty	Scenario	Components supplied by manufacturers can come under warranty.
Order	Scenario	Understocked components can be renewed through an order.
Progress Report	Scenario	Customers are notified of their problems' relative resolution via progress reports.
Normal Service	Scenario	Installations functioning as intended are operating at normal service.
Schedule	Scenario	Technicians' daily work is organized in a schedule.
Weather	Scenario	Changes like a woman's mood.

8.2: Candidate Classification

Candidate	Classification	Decision	Reason
Customer	Conceptual Class	Keep	Obvious.
Site	Conceptual Class	Keep	Obvious.
Access Method	Conceptual Class	Keep	Sites can have multiple.
Installation	Conceptual Class	Keep	Obvious.
Component	Conceptual Class	Keep	Obvious.
Manufacturer	Conceptual Class	Keep	Obvious.
Symptom Report	Conceptual Class	Keep	Separating these from work reports makes it easier to filter informatios from customers.
Fault	Conceptual Class	Keep	Obvious.

Candidate	Classification	Decision	Reason
Assessment	Conceptual Class	Keep	Can be separated into fault and resumption of normal service assessments, but I don't want to.
Visit	Conceptual Class	Keep	Obvious.
Technician	Conceptual Class	Keep	Obvious.
Skill	Conceptual Class	Keep	It can even be expanded but I don't want to do that.
Equipment	Conceptual Class	Keep	Like skills, but can be thrown.
Quotation	Conceptual Class	Keep	Obvious.
Approval	Transaction	Keep	It's safer, legally, to have its own class. I will not elaborate.
Work	Conceptual Class	Merge with work report	~~Might merge with work report.~~ It did.
Work Report	Conceptual Class	Keep	Might merge with work.
Inventory	Conceptual Class	Remove	Component and Equipment handle it well enough.
Assignment	Conceptual Class	Merge with component	I see it, but not on its own.
Reassignment	Conceptual Class	Remove	Assignment is sufficient.
Warranty	Conceptual Class	Merge with component	I see it, but not on its own.
Order	Conceptual Class	Keep	Might be doing something like this at my placement.
Progress Report	Attribute	Merge with work report	It'll work.
Normal Service	Conceptual Class	Merge with installation	It's one bool.
Schedule	Attribute	Merge with visit	It's two date-times.
Weather	Attribute	Remove	Not needed, unreliable.

8.3: Candidate List Refinement

Candidate	Decision	Final Term	Justification
Customer	Keep	Customer	Optimisation.
Site	Keep	Site	
Access Method	Keep	Access Method	
Installation / Normal Service	Merge	Installation	
Component / Warranty / Assignment	Merge	Component	
Manufacturer	Keep	Manufacturer	
Symptom Report	Keep	Symptom Report	
Fault	Keep	Fault	
Assessment	Keep	Assessment	
Visit / Schedule	Merge	Visit	
Technician	Keep	Technician	
Skill	Keep	Skill	
Equipment	Keep	Equipment	
Quotation	Keep	Quotation	
Approval	Keep	Approval	
Work / Work Report / Progress Report	Merge	Work Report	
Order	Keep	Order	

8.4: Domain Glossary

Candidate	Definition	Related	Evidence
Customer	Human service consumer.	Site, Quotation, Approval	Scenario, etc.
Site	Installation location.	Customer, Installation, Access Method	Scenario.
Access Method	Site entry protocol.	Site	Scenario, etc.

Candidate	Definition	Related	Evidence
Installation	Functional mess of components.	Component, Site, Fault	Scenario, etc.
Component	Parts of an installation.	Installation, Manufacturer, Fault	Scenario, etc.
Manufacturer	Component supplier.	Component	Scenario, etc.
Symptom Report	Notice of potential substandard operational efficiency.	Fault, Installation	Scenario, etc.
Fault	Measurable instance of substandard operational efficiency.	Installation, Compoent	Scenario, etc.
Assessment	Test of substandard operational efficiency.	Visit, Installation, Symptom Report	Scenario, etc.
Visit	Attempt at restoring standard operational efficiency.	Technician, Site	Scenario, etc.
Technician	Operator tasked with assessing, maintaining, and restoring standard operational efficiency.	Visit, Skill, Equipment, Work Report	Scenario, etc.
Skill	Abilities and knowledge utilised in assessment, maintenance, and restoration of standard installation operational efficiency.	Technician	Scenario, etc.
Equipment	Tools used in assessment, maintenance, and restoration of standard installation operational efficiency.	Technician	Scenario, etc.
Quotation	Cost of work done.	Customer, Work Report, Approval	Scenario, etc.
Approval	Agreement to cost of work done.	Quotation, Customer	Scenario, etc.
Work Report	Report describing work done.	Visit	Scenario, etc.
Order	Replenishment of component stock.	Component, Manufacturer	Scenario, etc.

9. Domain Model:

9.1: Selected Attributes

Candidate	Attributes	Reason
Customer	customerID, name, phoneNumber, email	No.
Site	siteID, name, address	
Access Method	accessMethodID, accessInfo	
Installation	installationID, model, type, condition	
Component	componentID, model, serialNumber, warranty, isInstalled, isReserved	
Manufacturer	manufacturerID, name, phoneNumber, email	
Symptom Report	symptomReportID, symptomDescription	
Fault	faultID, description, priority, status	
Assessment	assessmentID, description, resolution, status	
Visit	visitID, startTime, endTime	
Technician	technicianID, name	
Skill	skillID, name, proficiency	
Equipment	equipmentID, name	
Quotation	quotationID, price	
Approval	approvalID, signature, audioEvidence, videoEvidence	
Work Report	workReportID, tasks, observations, readings	
Order	orderID, price, orderDate, deliveryDate	

9.3: Associations and Multiplicity

Association	Class A Multiplicity	Class B Multiplicity	Justification
Customer - Site	1	0..*	
Site - Access Method	1	0..*	
Site - Installation	1	0..*	
Installation - Component	1..*	0..*	
Manufacturer - Component	1	1..*	
Installation - Symptom Report	1	0..*	
Symptom Report - Fault	0..*	0..*	
Site - Visit	1	0..*	
Visit - Technician	1..*	0..*	
Technician - Skill	0..*	0..*	
Technician - Equipment	0..*	0..*	
Customer - Quotation	1	0..*	
Quotation - Approval	1	1	
Visit - Work Report	1	1..*	
Work Report - Installation	0..*	1	
Installation - Assessment	1	0..*	
Component - Order	1..*	1	

9.4: Domain Model Diagram

See /diagrams/ca1-domain-model.puml

10. Traceability Matrix:

I still have no idea what this means. :(

11. Reflection & AI declaration:

11.1: Reflection

11.1.1: Important Decisions

I made the important decisions to merge work and work report and to stop coming up with justifications and reasons for decisions made on instinct.

11.1.2: Revisions

I revised the reason for the work candidate class to reflect its merger with the work report class.

11.1.3: Limitations

The major limitation was that I had little to no idea what I was doing.

11.1.4: Unresolved Questions

The main unresolved question is that I still have little to no idea what I'm doing. Please advise accordingly, as it will be greatly appreciated.

11.2: AI Declaration

I used AI to identify mechanical data storage mediums in order to eliminate them from consideration. I used it to remember the word consolidated. I used it to settle on the word inconsistent.